# User Study – ML2++ (Textual DSL)

*Instructions:* Fill Section A once. For Section B, answer for *each* use case you worked on. If you did multiple use cases, **duplicate Section B** per use case.

**Participant ID:** ____________ **Use case(s):** ☑ U1 River Flow ☐ U2 Smart-Home Energy ☐ U3 Solar Energy

## Section A – Background (answer once)

*This section helps us understand your experience level so we can analyze results fairly.*

**A1. Educational level** *(highest completed)*

*Tell us the highest level of education you have finished.*

☐ High School ☐ Vocational/Technical ☐ Bachelor's ☑ Master's ☐ Doctorate ☐ Other: ____________

**A2. Primary role** *(pick the closest)*

*Choose the option that best matches your current job or main activity. If none fits, select "Other".*

☐ IoT/CPS Developer ☐ MDE Practitioner ☐ Data Scientist / ML Engineer ☐ Embedded SW Engineer ☐ Educator / Researcher ☑ Software Architect / Full-stack ☐ System Integrator ☐ Early Adopter / Innovator ☐ Other: Devops

**A3. Self-rated expertise** *(tick one per row)*

*For each area, select your skill level. "IoT/CPS" = connected devices/systems. "MDE" = using models to guide or generate code. "ML/Data Science" = working with data and ML.*

| | Beginner | Intermediate | Advanced | Expert |
|---|---|---|---|---|
| IoT/CPS development | ☑ | ☐ | ☐ | ☐ |
| Model-Driven Engineering (MDE) | ☑ | ☐ | ☐ | ☐ |
| Machine Learning / Data Science | ☐ | ☑ | ☐ | ☐ |

**A4. Prior experience** *(used DSLs/modeling tools/ML frameworks before?)*

*Have you used similar tools (e.g., modeling editors, DSLs, ML libraries/frameworks)?* tensorflow

☑ Yes ☐ No

## Section B – Task-Specific (duplicate this page *per use case*)

**This B-section is for:** ☑ U1 River Flow ☐ U2 Smart-Home Energy ☐ U3 Solar Energy

*If you did more than one use case, copy this whole Section B for each one and fill it separately.*

## B1. Time & Effort

*Please record your times honestly. If unsure, give your best estimate. "Extra learning time" = time reading docs/tutorials to figure things out.*

### B1. Time spent (minutes)

*Minutes you spent defining/editing the DSL, generating code, reviewing outputs, and any extra learning time.*

| | **ML2++ (Textual DSL)** |
|---|---|
| Defining DSL + generate + review | 60min |
| Extra learning time | 45min |

### B2. Statements written (DSL) *(exclude blanks/comments)*

*Approximate number of DSL lines/statements you actually typed or edited.*

| | **ML2++ (Textual DSL)** |
|---|---|
| ML2++ DSL statements | 156 |

## B2. Task Completion & Ease of Use

*Tell us if you finished the task and how easy/hard each step felt. Then write how many minutes you spent per step.*

### B3. Were you able to finish the use case?

*"None" = could not complete; "Partial" = finished some parts; "Complete" = finished everything.*

☐ None ☐ Partial ☑ Complete

### B4. Ease by workflow step (with per-step time)

*Rate difficulty (1=very easy, 5=very difficult) and write minutes spent. "Model configuration" = choosing algorithm and hyperparameters. "Model evaluation" = checking results with plots/metrics.*

| Step | 1 | 2 | 3 | 4 | 5 | **Time** |
|---|---|---|---|---|---|---|
| Whole workflow (end-to-end) | ☐ | ☐ | ☑ | ☐ | ☐ | ___ |
| Data preprocessing | ☐ | ☐ | ☑ | ☐ | ☐ | ___ |
| **Model configuration** (algorithm, hyperparameters) | ☐ | ☑ | ☐ | ☐ | ☐ | ___ |
| Model training | ☐ | ☑ | ☐ | ☐ | ☐ | ___ |
| Model evaluation (metrics/plots/inspection) | ☑ | ☐ | ☐ | ☐ | ☐ | ___ |

*Tip: use a simple stopwatch and pause for breaks. If unsure, estimate.*

### B5. Ease of ML2++ elements *(1=Very easy → 5=Very difficult)*

*Rate how easy it was to work with each part of the language/editor.*

| Element | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| Data types & objects | ☐ | ☑ | ☐ | ☐ |
| Things & behaviours (statecharts) | ☐ | ☑ | ☐ | ☐ |
| Ports & properties | ☐ | ☑ | ☐ | ☐ |
| **Model configuration options (algorithm/hyperparameters)** | ☐ | ☑ | ☐ | ☐ |
| Configuration (instances/connectors) | ☐ | ☑ | ☐ | ☐ |

## B3. Learning & Usability

*These questions focus on how quickly you learned the tool and how natural it felt.*

**B6. Learning curve** *(1=Very easy → 5=Very difficult)*

*How much effort did it take to learn enough to finish this use case?*

☐1 ☐2 ☑3 ☐4 ☐5

**B7. Intuitiveness** *(1=Very intuitive → 5=Not intuitive)*

*Did the tool feel natural/obvious to use, or did it feel confusing?*

☐1 ☐2 ☑3 ☐4 ☐5

**B8. Hardest part to learn** *(briefly)*

*Name the single most confusing or time-consuming thing to learn.*

learn how it work

## B4. Debugging & Error Handling

*Tell us about problems you faced and how hard they were to fix.*

**B9. Error frequency** *(0=None, 1=Few, 2=Moderate, 3=Many)*

*How many problems blocked you (errors, failed runs, missing packages, wrong paths, etc.)?*

☐0 ☐1 ☑2 ☐3

**B10. Clarity of error messages** *(1=Not clear → 5=Very clear)*

*When something went wrong, did the messages help you understand the fix?*

☐1 ☑2 ☐3 ☐4 ☐5

**B11. Debugging effort** *(1=Easy → 5=Hard)*

*Overall, how hard was it to fix your issues?*

☐1 ☑2 ☐3 ☐4 ☐5

**B12. Debugging time (minutes)** *(estimate)*

*About how many minutes did you spend fixing problems (total for this use case)?*

20 min

### B5. Code Quality, Maintainability, Flexibility

*These questions refer to the generated code you viewed and how easy it would be to change things later.*

**B13. Generated code quality** *(1=Poor → 5=Excellent)*

*How good is the generated code (readable, well-structured, correct)?*

☐1 ☐2 ☐3 ☑4 ☐5

**B14. Maintainability** *(1=Very difficult → 5=Very easy)*

*If you had to update/change your solution later (e.g., new dataset, new horizon), how easy would it be?*

☐1 ☐2 ☑3 ☐4 ☐5

**B15. Flexibility / Customisation** *(1=None → 5=Very flexible)*

*How easily can you change models, hyperparameters, thresholds, or add steps without workarounds?*

☐1 ☐2 ☐3 ☑4 ☐5

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## Section C – Overall Assessment (ML2++ only)

*Your overall opinion and which features helped you most.*

**C1. Overall satisfaction** *(1=Very dissatisfied → 5=Very satisfied)*

*Your overall feeling about using ML2++ for this use case.*

☐1 ☐2 ☐3 ☐4 ☑5

**C2. Would you use ML2++ for your next similar project?**

*If you had a similar project tomorrow, would you choose this approach? Tell us briefly why.*

☑ Yes ☐ No ☐ Undecided *Why?* beacuse you can focus on your problem and design our model better

**C3. Feature effectiveness** *(1=Ineffective → 5=Very effective)*

*How useful were these features for you?*

| Feature | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Time-series forecasting models | ☐ | ☐ | ☐ | ☑ | ☐ |
| Automatic preprocessing & plots | ☐ | ☐ | ☑ | ☐ | ☐ |
| Training configuration options | ☐ | ☐ | ☑ | ☐ | ☐ |
| Multi-day forecasting accuracy | ☐ | ☐ | ☑ | ☐ | ☐ |

**C4. Favourite feature:** define our models

*What was the single most helpful feature?*

**C5. Least favourite feature:** preprocessing

*Which feature felt least helpful or got in your way?*

**C6. Missing features you would like:** design to have better syntax i think yaml file its good

*What do you wish the tool had (e.g., a plot, algorithm, wizard, or shortcut)?*

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## Section D – Open Feedback

*Tell us anything else that would help us improve.*

**D1. Barriers to adoption** *(what would stop you from using ML2++ in production?)*

*Examples: setup, missing plots, docs, integration, performance, governance.*

low usage on community and lack of documentaction

**D2. One improvement you would prioritize**

*If we could fix/add only one thing, what would help you the most?*

support modular
use configuration file like yaml
try lower conf code

*Scale anchors used throughout for consistency:*

- Ease / Learning: 1=Very easy, 2=Easy, 3=Moderate, 4=Difficult, 5=Very difficult.
- Intuitiveness: 1=Very intuitive, 2=Intuitive, 3=Neutral, 4=Somewhat confusing, 5=Not intuitive.
- Error clarity: 1=Not clear, 2=Somewhat clear, 3=Clear, 4=Very clear, 5=Extremely clear.
- Code quality: 1=Poor, 2=Fair, 3=Good, 4=Very good, 5=Excellent.
- Maintainability: 1=Very difficult, 2=Difficult, 3=Moderate, 4=Easy, 5=Very easy.
- Flexibility: 1=None, 2=Limited, 3=Some, 4=Flexible, 5=Very flexible.
- Feature effectiveness: 1=Ineffective, 2=Low, 3=Moderate, 4=Effective, 5=Very effective.